SOFTWARE TEST PLAN (STP)

VWO – Digital Experience Optimization Platform

Target Application: https://app.vwo.com/

Document Reference:	VWO-STP-2026-V1.0
Product / Platform:	VWO Experience Optimization Platform (https://app.vwo.com/)
Prepared By:	QA Engineering & Automation Team
Product Manager & Reviewer:	Pramod Dutta (Principal PM, VWO)
Target Release Date:	October 2026 (Baseline PRD: January 7, 2026)

1. Document Control & Revision History

Version	Date	Author	Description	Status
1.0	Oct 2026	QA Team / Pramod Dutta	Production-ready Test Plan based on PRD	Approved

2. Objective

The primary objective of this Test Plan is to provide an enterprise-grade quality assurance framework for the **VWO (Visual Website Optimizer)** Digital Experience Optimization (DXO) and Conversion Rate Optimization (CRO) Platform. This plan guarantees functional correctness, statistical accuracy via Bayesian SmartStats, sub-50ms client delivery without page flicker, high-fidelity behavioral analytics (heatmaps, recordings), and enterprise data privacy compliance (GDPR/CCPA/2FA).

3. Scope of Testing

Testing Dimension	Detailed Scope & Verification Focus
Functional Testing	Full verification of A/B, Split URL, MVT campaigns, Visual/Code editor, SmartStats, Heatmaps, Recordings, Audience Segmentation, and Integrations.
Data & Statistics	Bayesian SmartStats probability calculations, conversion distributions, false discovery rate guardrails, and event tracking accuracy.
Performance & Latency	Web application workflows respond within 2.0s; tracking snippet executes in <50ms with zero perceptual page flicker.
Security & Governance	2FA, SAML SSO, Role-Based Access Control (Admin, Marketer, Designer, Viewer), OWASP Top 10 protection, and audit logging.
Privacy & Compliance	GDPR, CCPA adherence, cookie consent enforcement, and automated PII keystroke masking in session recordings.
Cross-Browser/Device	Chrome, Firefox, Safari, Edge across Windows 11/10, macOS Sonoma, iOS 17/18, and Android 13/14.
Integrations	Event forwarding and sync with Google Analytics 4, Mixpanel, Segment CDP, Snowflake, Salesforce, and Shopify.

4. Inclusions: Feature-by-Feature CRUD & Workflow Matrix

Feature / Module	CRUD Operations & Key Workflows	Acceptance Criteria
A/B, Split & MVT Testing	Create, preview, schedule, launch, pause, and conclude experiments across multi-variations.	Variations render faithfully; traffic split allocations strictly adhere to configured ratios.
SmartStats Engine	Calculate Bayesian conversion distributions, chance to beat control, and credible intervals.	Accurate statistical calculations; alerts trigger when confidence thresholds are achieved.
Visual & Code Editor	WYSIWYG point-and-click edits; custom CSS/JS injection; DOM element manipulation.	Edits apply seamlessly in preview and production without breaking host page scripts.
Heatmaps & Insights	Click maps, scroll depth percentages, and mouse movement/focus tracking.	Density overlays match visitor interactions; scroll markers align with responsive breakpoints.
Session Recordings	Capture visitor sessions with variable playback speed and inactivity skipping.	100% PII masking on input fields, passwords, credit card inputs; no dropped sessions.
Audience Targeting	Segment rules based on Geo, device, referral URL, cookies, and custom JS variables.	Matched visitors receive targeted variation; unmatched visitors receive control.
Personalization Engine	Deliver tailored real-time experiences and customized dynamic CTAs to visitor segments.	Dynamic content swap occurs instantaneously (<50ms) without page flicker.
Integrations	Connectors for Google Analytics 4, Mixpanel, Segment, Shopify, Salesforce.	Bidirectional sync verified; custom dimensions and event parameters populated accurately.

5. Test Environments & Defect Triage Team

Role / Area	POC Name	Core Responsibilities
QA & Automation Lead	Divya / QA Lead	Test plan ownership, test execution, automation authoring, defect triage
Product Management	Pramod Dutta	Requirement clarification, scope prioritization, final launch sign-off
Frontend Engineering	Devesh	Visual & Code editor, web console workflows, client-side rendering
Backend & Data Science	Sonal	SmartStats calculations, ingestion pipelines, REST API services
DevOps & Cloud Infra	Prajeeth	Environment uptime, CI/CD pipelines, edge CDN snippet delivery

6. Test Schedule & Milestones

Sprint / Phase	Milestone / Activities	Timeline	Owner
Phase 1	PRD Review, Test Plan Authoring & Sign-Off	Sprint 1 (Days 1–3)	QA Lead / Pramod Dutta
Phase 2	Test Scenario & Test Case Design	Sprint 1 (Days 4–5)	QA Team
Phase 3	Execution Cycle 1 (Core Functional Suites)	Sprint 2 (Days 1–5)	QA Team
Phase 4	Defect Verification, Retesting & Cycle 2	Sprint 3 (Days 1–3)	QA / Dev Leads
Phase 5	Non-Functional QA (Performance, Security, Cloud QA)	Sprint 3 (Days 4–5)	Security & QA Leads
Phase 6	Final Regression, Summary Report & Go/No-Go Decision	Sprint 4 (Days 1–2)	Pramod Dutta & Stakeholders

7. Risks & Mitigations

Risk Description	Severity	Mitigation Strategy
Page Flicker (FOOC)	High	Rigorously validate asynchronous anti-flicker snippet under artificial network latency.
Analytics Discrepancy	High	Execute dual-tracking reconciliation suites ensuring discrepancy stays within ±3% tolerance.
High Traffic Spikes	Critical	Perform load testing up to 50k req/s; implement resilient message queuing and auto-scaling.
Third-Party AdBlockers	Medium	Validate First-Party Custom Domain Proxying configuration and fallback tracking.

8. Approvals & Sign-Off Matrix

Role	Stakeholder Name	Designation	Approval Status	Date
QA Lead	Divya	Lead QA Specialist	Approved	October 2026
Product Manager	Pramod Dutta	Principal PM (VWO)	Pending Final Review	October 2026
Frontend Lead	Devesh	Lead Frontend Architect	Approved	October 2026
Backend Lead	Sonal	Lead Backend Architect	Approved	October 2026
DevOps Lead	Prajeeth	Lead DevOps Engineer	Approved	October 2026